

Avishai Kolet - Electrical Engineer

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Hardware Design Engineer

B.Sc. Electrical Engineering graduate with strong foundations in digital/analog design and VLSI. Gained extensive hands-on experience in advanced VLSI and FPGA labs while mentoring students as a private tutor. Fast learner with proven debugging skills, clear technical communication, and a passion for turning complex designs into working silicon.

Experience

2025-2024 SoC Design for Hardware Reliability for Bar-Ilan University

Designed a Hardware reliability data acquisition system for an undergraduate lab

- Designed an overlay using Verilog on PYNQ-Z2 board based on the Zynq-7020 FPGA, using Vivado design suite.
- Created a system to automate reliability test runs based on python scripts, remotely control and monitor a SP540 oven controller and running Jupyter-notes from the PYNQ board.

Private Projects

2013 - 2025 Private Tutor: For Undergraduate and high school level courses

2023 Intel CPUgarage: Learned hardware coding of a simple retro game using Verilog on de10-lite Altera based FPGA

2023 NeuroHack: Programmed a PTSD treatment mobile game using Unity

2022 Hackathon: Made a flowrate estimator using Arduino

Education

2025 graduated B.Sc. Electrical Engineering at Bar-Ilan University - GPA 83.07

Major 1: Nanoelectronics and VLSI

Major 2: Electro-Optics

Final Project: Design of efficient configurable systolic array processing element for Machine Learning using Verilog and the university provided development suits and technologies

Skills

Digital Design & Verification: SystemVerilog, Verilog, ModelSim

RTL-to-GDSII Flow & Cadence Tools: Genus (Synthesis), Innovus (Place & Route), Voltus (Power), Virtuoso (Custom Layout), PVS, Quantus

FPGA Prototyping & Tools: Xilinx Vivado, Intel Quartus Prime

Programming & Scripting: C/C++, Python, Tcl, Bash

Simulation & Analog/Mixed-Signal: Spectre, Xcelium, SimVision, MATLAB/Simulink

Other: Git, Linux, Static Timing Analysis, DRC/LVS

Languages

English & Hebrew - Native Language

Hindi & Marathi - Good Level

Military Service

2014 - 2017 Served as a fighter in an Artillery battalion